



Pelita Bangsa
Academy

Building Frontend UI for Web3 DApp

BUILDing with IDRX Stablecoin on Lisk Bootcamp

Hi, I'm **Stefanus Deo** from Indonesia 

4+ years

Experience in the IT industry
stefanusdeo.site



**Universitas Multimedia
Nusantara**

S.Kom. | Information Technology

Working Experience

(Startup, Corporate, Freelance)



Teaching Experience



Pelita Bangsa Academy

**Mall of Indonesia, Lantai 2, Unit 2F-A7B,
Jalan Boulevard Barat Raya No. 1,
Kelapa Gading Jakarta Utara 14240.**

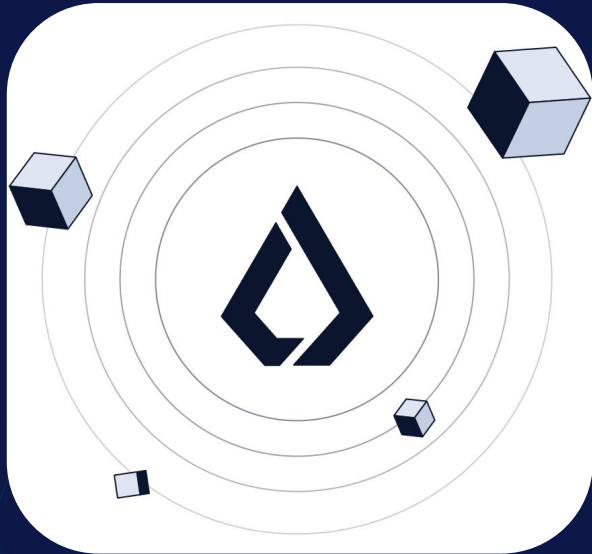


”

Upskill dan Reskill Talenta Teknologi Indonesia di bidang Blockchain sehingga kalian bisa bersaing secara global.



Lisk Blockchain



- Founded in 2016 by Max Kordek, and Oliver Beddows
- Part of [Ethereum ecosystem](#)
- A Growing [EVM Layer 2](#) (L2) Chain
- Part of [Optimism Superchain](#)
- [Near zero](#) fees

SESI

01 Introduction to Lisk Blockchain & IDRX Stablecoin

02 Smart Contract Development on Lisk

03 Interacting with IDRX Stablecoin

SESI

04

Building Frontend UI for Web3 DApp

HARI INI

**Extra
Class**

Introduction to Xellar

17 Apr

05

Integrating Xellar Kit with Smart Contract

23 Apr

06

Brainstorming Web3 Project Ideas

30 Apr

Agenda

**Goal of this
Bootcamp**

Materi

Hands On

Q&A

Wrap Up

Foto Bersama



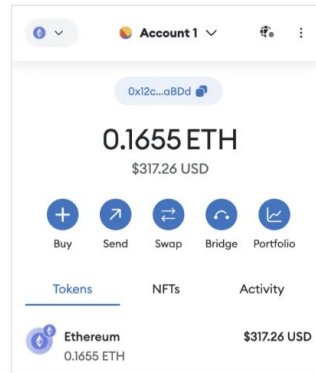
Prepare for Hands On

Chrome

iOS

Android

Install MetaMask for your browser



<https://metamask.io/download/>

Recommendation



Physical Notebook



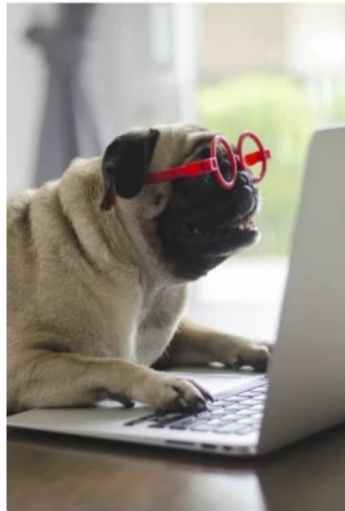
<https://notion.com>

Be Active

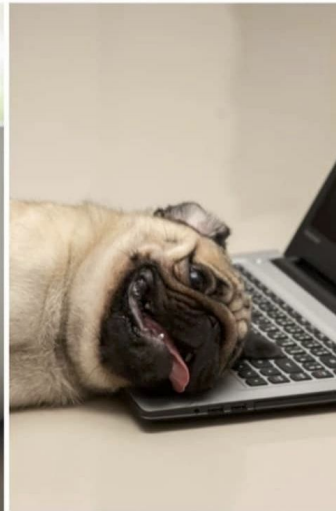


Open Your Camera

Zoom meeting,
with video



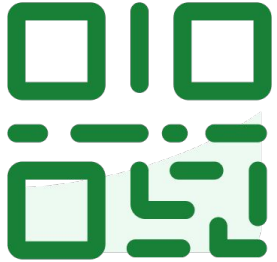
Zoom meeting,
audio only



Prepare for Hands On

1. Visual Studio Code
<https://code.visualstudio.com/download>
2. Node
<https://nodejs.org/en/download/prebuilt-installer>

slido



Join at slido.com
#9779256



Click **Present with Slido** or install our
instructions for participants while presenting.

to display joining

A

Goal of this Bootcamp



Pelita Bangsa Academy Lisk 1000 IGNITE
BlockDevId X IDR X Xellar

* ♦ *

BUIDLing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

The goal of this bootcamp:

- Learn about Lisk Blockchain Technology
- Come up with Web3 Application Ideas
- #BuildOnLisk Blockchain
- Participate Lisk Builders Challenge Round 1
- Build successful Web3 Startup !!



Fund Tracking Management





Lisk

BlockDevId



IDRX



Xellar



Lisk

Builders
Round 1
Challenge

bit.ly/LBC-Round-1



Project Requirement:

1. Smart Contract deployed on **Lisk Sepolia Testnet** or **Lisk Mainnet** and verified on **Blockscout Explorer**
2. Integrate your dApp with **IDRX Stablecoin** on **Lisk Mainnet** or **Lisk Sepolia Testnet**
3. Build your dApps with **Xellar Kit**
4. Live Project URL

IDRX Address on Lisk Mainnet: 0x18Bc5bcC660cf2B9cE3cd51a404aFe1a0cBD3C22

IDRX Address on Lisk Sepolia Testnet: 0xD63029C1a3dA68b51c67c6D1DeC3DEe50D681661

Prize:

3,000 USD (1st Winner)

(32 Million IDR^X & USD 1,000 Xellar Credits)

1,875 USD (2nd Winner)

(22 Million IDR^X & USD 500 Xellar Credits)

1,050 (3rd Winner)

(12 Million IDR^X & USD 300 Xellar Credits)

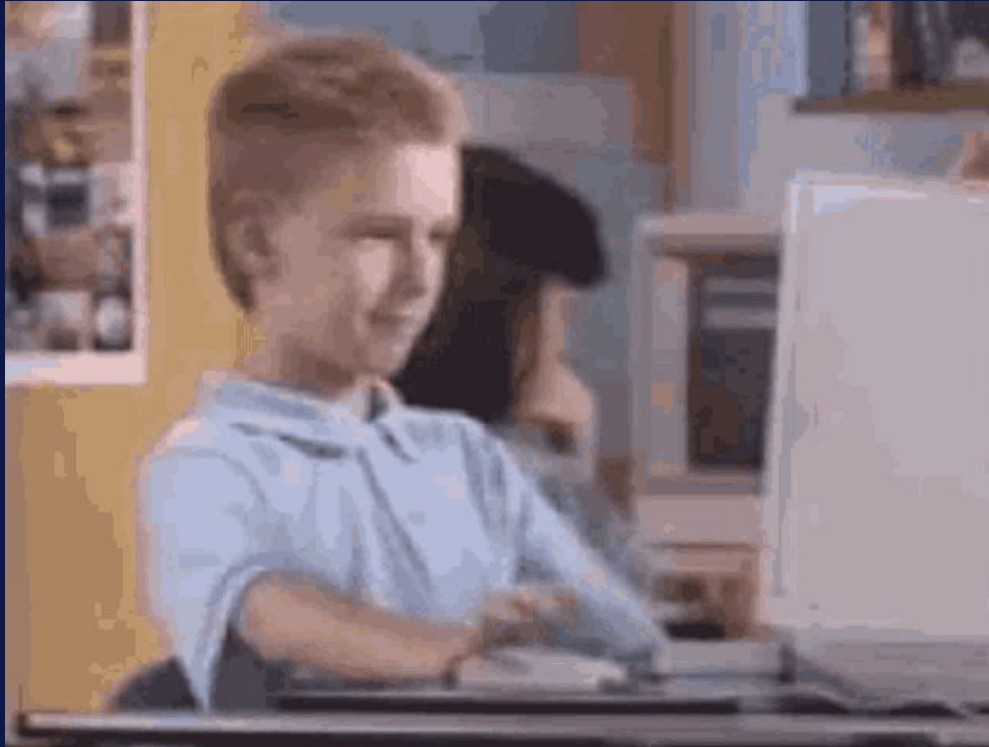


Here's what the winners get

1 st winner	3000 ^{USD} (32MIO IDR ^X & USD 1,000 Xellar Credits)
2 nd winner	1875 ^{USD} (22MIO IDR ^X & USD 500 Xellar Credits)
3 rd winner	1050 ^{USD} (12MIO IDR ^X & USD 300 Xellar Credits)

Challenge

Not bad



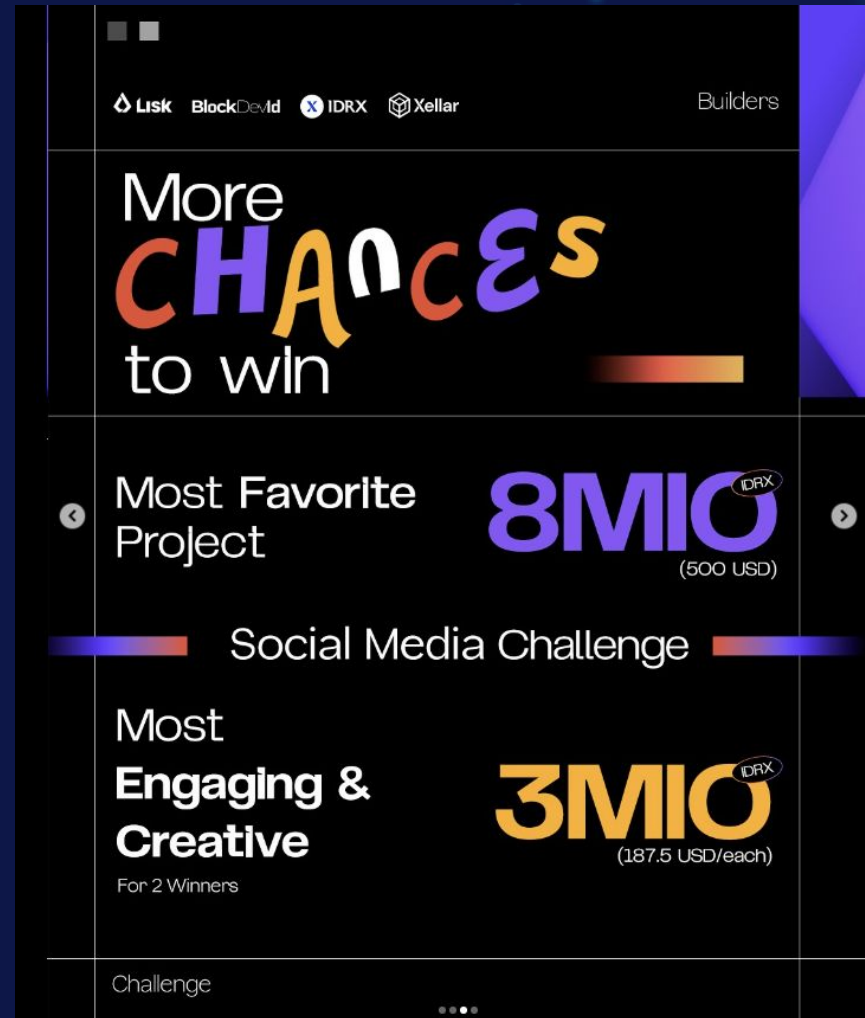


Prize:

8 Million IDR^X (Most Favorite Project)

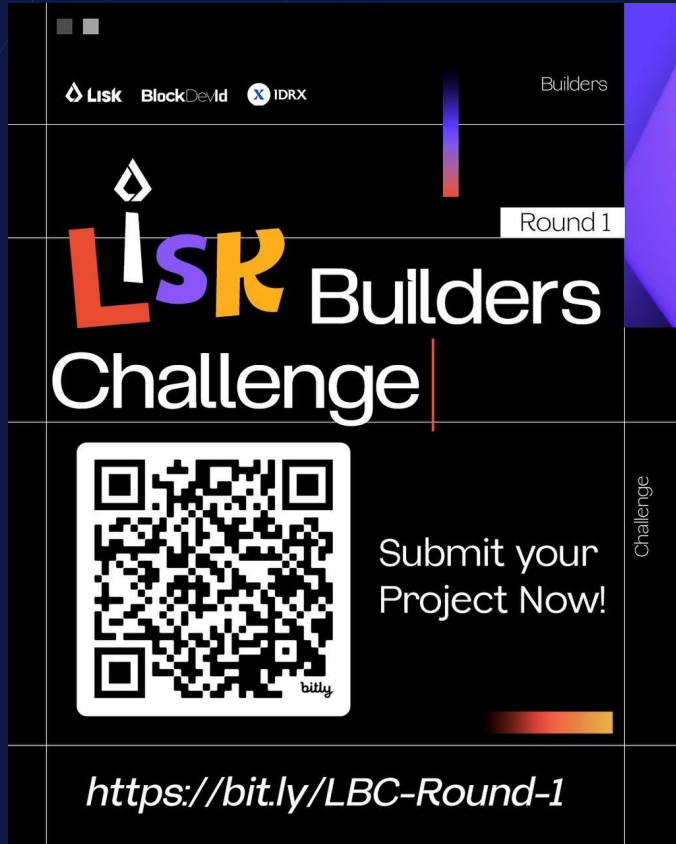
3 Million IDR^X (SocMed Challenge Winner #1)

3 Million IDR^X (SocMed Challenge Winner #2)



YESSSS!!





The poster for the Lisk Builders Challenge Round 1 features a dark background with a purple and blue gradient on the right side. At the top left, there are logos for Lisk, BlockDevId, and IDRX. The word "Builders" is in the top right. The main title "Lisk Builders Challenge" is prominently displayed in the center, with "Lisk" in large, colorful letters (red, purple, orange) and "Builders Challenge" in white. Below the title is a large QR code with the text "Submit your Project Now!" to its right. At the bottom, the URL <https://bit.ly/LBC-Round-1> is shown. A vertical label "Challenge" is on the right edge of the poster.

Timeline:

- Hackathon Start: March 8, 2025
- Kickoff AMA: March 12, 2025
- Workshop: March 19 – April 30, 2025
- Submission Deadline: May 17, 2025
- Announcement of Finalists: May 24, 2025
- Finalists Coaching: May 26–30, 2025
- Demo Day (Jakarta – Hybrid): May 31, 2025

Lisk Developer Experience Survey Form



<https://forms.gle/bMj56NQwCci3aSPR8>



B

Pelita Bangsa Academy Resources



Pelita Bangsa Academy Lisk 1000 IGNITE
BlackDevid X IDR X Xellar

* ♦ *

BUILDing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

Pelita Bangsa Academy Resources:

- **LMS (Learning Management System):**
<https://lms.pelitabangsa.co.id/>
(Class Recording, PDF Material & Other Links)
- **Discord Link:**
<https://discord.gg/Jquvw5DqEv>
(Discussion on #lisk channel)
- **Zoom Links on Google Calendar:**
<https://calendar.google.com/calendar/embed?src=0028bfb69c6aa9e82732209167a85aabf5201caae48f1653782c0365b238277%40group.calendar.google.com&ctz=Asia%2FJakarta>

IDRX on Lisk Sepolia Airdrop Form



<https://forms.gle/t5NSNTdSRWEgsuDMA>

Lisk Sepolia OP Superchain Faucet

Superchain Faucet

Get free testnet tokens for building applications on the Superchain.

Sign in to use the faucet

Anyone can claim 0.1 test ETH every 24 hours, or verify your onchain identity for more tokens. [See details](#)

Sign in

Network

 OP Sepolia	 Base Sepolia
 Unichain Sepolia	 Ink Sepolia
 Minato Sepolia	 Worldchain Sepolia
 Zora Sepolia	 Mode Sepolia
 Lisk Sepolia	 Cyber Sepolia

<https://console.optimism.io/faucet>

Lisk Sepolia PBA Faucet

Lisk Sepolia Testnet Faucet

Request Sepolia ETH for testing on the Lisk Sepolia Testnet

Chain Overview

+ Add to MetaMask

Network Name	Lisk Sepolia
RPC URL	https://rpc.sepolia-api.lisk.com
Chain ID	4202
Currency Symbol	ETH
Block Explorer	https://sepolia-blockscout.lisk.com

Request Testnet ETH

Connected: 0x47331edc7220ad93d62130ce64c10f7166f4c947

Request 0.01 ETH to Connected Wallet

<https://lisk-sepolia-faucet.vercel.app/>

00

About Danantiri

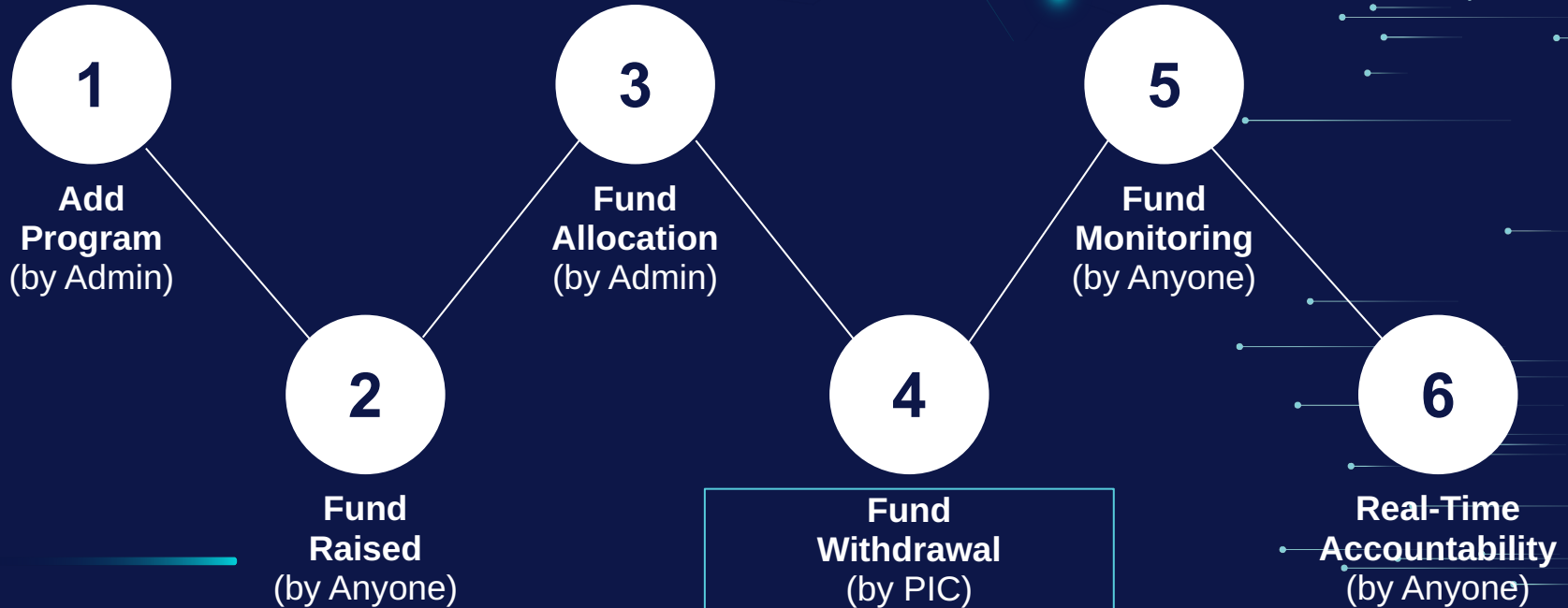
The Problem

- Transparency Issues
- Corruption Risk
- Delayed Reporting

Solution - Danantiri

- Immutable Record
- IDRX stablecoin
- Transparent
- Real-Time Accountability

Business Process



Danantiri | Top Secret



Support Meaningful Projects with Crypto

Your allocated fund will be pooled and distributed to impactful programs.

[Contribute](#)

Managed Fund Amount

12.5 IDR

Programs Funded

4

Featured Programs

MBG

Biar anak sekolah makan enak

0x1RGT7154748V1CE5176479CD0B574AE2448939B5

Testing Program

Testing Program

0x1473289472389578239dfjes

Pembangunan Jalan Tol

Pembangunan Jalan Tol untuk Kontainer

0x3432497DFUOWERN343248098

Pembangunan Sekolah

Pembangunan Sekolah Rakyat

0x3423U43IO24N3B2340934857

Danantiri

Blockchain-Based Government Fund Tracking

Supported Cryptocurrencies

💎 IDRX Lisk Sepolia Testnet

Connect With Us

🗨 Discord PBA

Danantiri | Top Secret

Contribute to Danantiri

Enter the amount you want to contribute.

Enter amount in IDR

Approve IDR

Contribute

Danantiri | Top Secret

[Connect Wallet](#)

Admin Panel - Fund Programs Management

Add New Program

[Add Program](#)

Current Programs

MBG

Biar anak sekolah makan enak

0x1RG77154748V1CE5176479CDG8574AE244B939B5

Testing Program

Testing Program

0x1473289472389678239df/sas

Danantiri | Top Secret

×

MBG

Biar anak sekolah makan enak

PIC Address:
0x1RGT7154748V1CE5176479CDGB574AE244B939B5

Allocate

Remaining Fund:

0 IDR / 25,000,000,000 IDR

Cancel



Danantiri | Top Secret

Danantiri | Top Secret

Pembangunan Sekolah

Pembangunan Sekolah Rakyat

PIC Address: 0x3423U43IO24N3B234G934857

Remaining Fund:

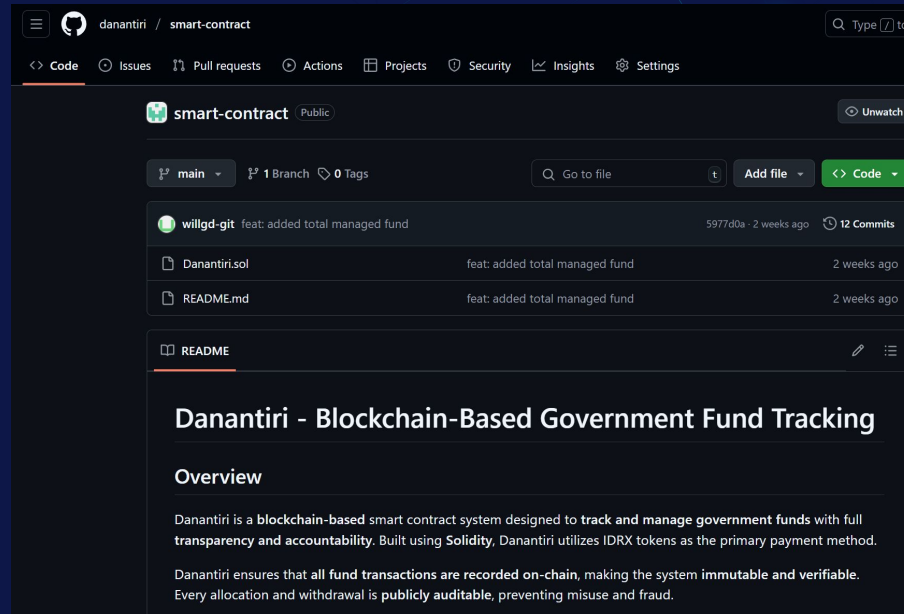


100,000,000 IDR / 130,000,000 IDR

Withdraw History

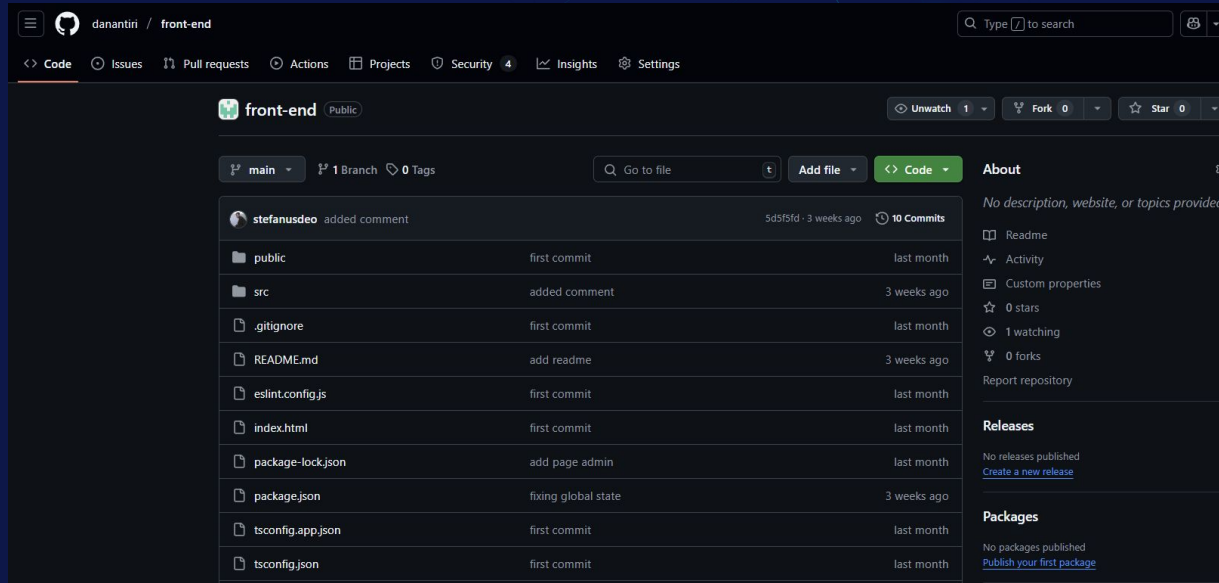
Date	Amount	Note
30000000	25-03-2025	Pembayaran Kontraktor

GitHub Repo - Smart Contract



<https://github.com/danantiri/smart-contract>

GitHub Repo - Front End



The screenshot shows the GitHub interface for the repository 'danantiri / front-end'. The repository is public and has 1 branch (main) and 0 tags. The file list includes:

File	Commit Message	Time
public	first commit	last month
src	added comment	3 weeks ago
.gitignore	first commit	last month
README.md	add readme	3 weeks ago
eslint.config.js	first commit	last month
index.html	first commit	last month
package-lock.json	add page admin	last month
package.json	fixing global state	3 weeks ago
tsconfig.app.json	first commit	last month
tsconfig.json	first commit	last month

The right sidebar shows the 'About' section with no description, website, or topics provided. It also lists 'Releases' (no releases published) and 'Packages' (no packages published).

<https://github.com/danantiri/front-end>

01

Frontend Recap

SPA FRAMEWORK / LIBRARY

Single Page Application (SPA) is a web application that does not use the default method of loading new pages completely.



ALTERNATIVE OF SPA



NEXT.js



NUXTJS

SSR : Server-side rendering
SSG : Static Site Generation

CSS Frameworks





Learn Further

1. HTML (3 jam 42 menit)

<https://youtu.be/hMDJyb7VkYw>

(Programmer Zaman Now)

2. CSS (3 jam 29 menit)

<https://youtu.be/8FhmFuGLhaQ>

(Programmer Zaman Now)

3. Javascript (8 jam 3 menit)

https://youtu.be/SDROba_M42g

(Programmer Zaman Now)



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Learn Further

1. React

<https://youtube.com/playlist?list=PLFIM0718LjIUu3X2zYNqomEWS3sYd-fV1>

(Web Programming UNPAS)

2. Bootstrap

<https://youtube.com/playlist?list=PLFIM0718LjIWizLOiZFlaXXblszkWqIPR>

(Web Programming UNPAS)

3. Tailwind CSS

https://youtube.com/playlist?list=PLFIM0718LjIUHFRMzPJ0wGjx9_NIC5d1h

(Web Programming UNPAS)

4. Shadcn

<https://youtu.be/ORUImQXulmAL>

(VoidFnc)



**Pelita Bangsa
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02

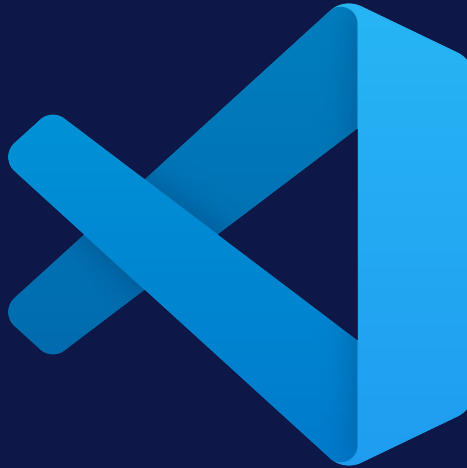
Installation

Installation Node Js



<https://nodejs.org/>

Installation VSCode



<https://code.visualstudio.com/download>

Install React (Vite)

```
[→ PBA npm create vite@latest first-project
Need to install the following packages:
create-vite@6.3.1
Ok to proceed? (y) y

> npx
> cva first-project

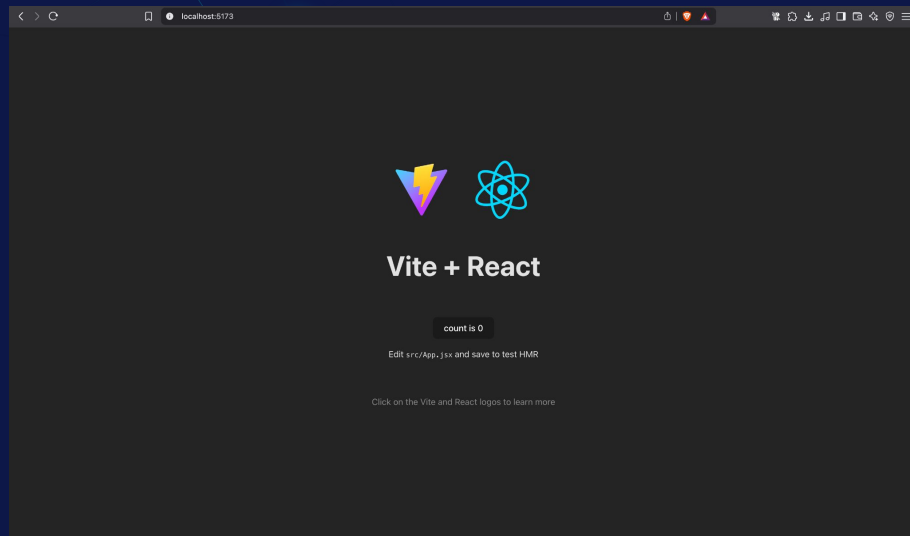
|
| Select a framework:
|   React
|
| Select a variant:
|   JavaScript
|
| Scaffolding project in /Users/stefanusdeo/Documents/Work/Partime Work/PBA/first-project...
|
| Done. Now run:
|
| cd first-project
| npm install
| npm run dev
|
npm notice
npm notice New major version of npm available! 10.9.2 -> 11.3.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.3.0
npm notice To update run: npm install -g npm@11.3.0
npm notice
→ PBA █
```

<https://vite.dev/guide/>

Install React (Vite)

```
[→ PBA cd first-project  
→ first-project npm install
```

```
first-project npm run dev
```



<https://vite.dev/guide/>

Structure Folder

```
▼ FIRST-PROJECT
  > node_modules
  > public
  ▼ src
    ▼ assets
      * react.svg
      App.css
      App.jsx
      index.css
      main.jsx
    .gitignore
    eslint.config.js
    index.html
    package-lock.json
    package.json
    README.md
    vite.config.js
```

<https://vite.dev/guide/>

03

Introduction React

03.A

What Is .JSX?

```
<!DOCTYPE html>

<html>

<title>HTML Tutorial</title>

<body>

<h1>This is a heading</h1>

<p>This is a paragraph</p>

</body>

</html>
```

```
import { useState } from 'react' 4.6k (gzipped: 1.9k)
import reactLogo from './assets/react.svg'
import viteLogo from '/vite.svg'
import './App.css'

function App() {
  const [count, setCount] = useState(0)

  return (
    <div>
      <a href="https://vite.dev" target="_blank">
        <img src={viteLogo} className="logo" alt="Vite logo" />
      </a>
      <a href="https://react.dev" target="_blank">
        <img src={reactLogo} className="logo react" alt="React logo" />
      </a>
    </div>
    <h1>Vite + React</h1>
    <div className="card">
      <button onClick={() => setCount((count) => count + 1)}>
        count is {count}
      </button>
      <p>
        Edit <code>src/App.jsx</code> and save to test HMR
      </p>
    </div>
    <p className="read-the-docs">
      Click on the Vite and React logos to learn more
    </p>
  </>
)
```

Code

```
const element = <h1>Hello, React!</h1>;
```

```
const name = "Deo";
```

```
const element = <p>Halo, {name}!</p>;
```

03.B

Component

```
function Greeting() {  
  return <h1>Halo, saya belajar React!</h1>;  
}
```



```
import Greeting from './Greeting';
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
    <h1>Halo, saya belajar React!</h1>
```

```
    <h1>Halo, saya belajar React!</h1>
```

```
    <h1>Halo, saya belajar React!</h1>
```

```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```

```
import Greeting from './Greeting';
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
      <Greeting />
```

```
      <Greeting />
```

```
      <Greeting />
```

```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```

03.C

Props

```
function Greeting(props) {  
  return (  
    <h1>Halo, Saya {props.name}, saya ingin belajar React!</h1>  
  );  
}
```

```
import Greeting from './Greeting';

function App() {
  return (
    <div>
      <h1>Halo Deo, saya ingin belajar React!</ h1>
      <h1>Halo Jason, saya ingin belajar React!</ h1>
      <h1>Halo Budi, saya ingin belajar React!</ h1>
    </div>
  );
}

export default App;
```

```
import Greeting from './Greeting';

function App() {
  return (
    <div>
      <Greeting name="Deo" />
      <Greeting name="Jason"/>
      <Greeting name="Budi"/>
    </div>
  );
}

export default App;
```

03.D

State

```
function Counter () {  
  const [count, setCount] = useState(0);  
  
  return (  
    <div>  
      <p>Nilai sekarang: {count}</p>  
      <button onClick={() => setCount(count +  
1)}>Tambah</button>  
    </div>  
  );  
}
```

```
const [count, setCount] = useState(0);
```

```
//count = show value
```

```
//setCount = update value
```

```
const [name, setName] = useState('');
```

```
const [old, setOld] = useState(false);
```

03.E

Event Handling

```
function Counter() {  
  const [count, setCount] = useState(0);  
  const handleAdd = () => {  
    const temp = count + 1;  
    setCount(temp);  
  }  
  
  return (  
    <div>  
      <p>Nilai sekarang: {count}</p>  
      <button onClick={handleAdd}>Tambah</button>  
    </div>  
  );  
}
```

03.F

Conditional Rendering

```
function App() {  
  const [old, setOld] = useState(false);  
  return (  
    <div>  
      {old ? <p>Wahh Saya sudah tua</p> : <p>Saya masih muda</p>}  
    </div>  
  );  
}
```

03.G

Hooks Dasar

```
import { useEffect } from 'react';
```

```
useEffect(() => {  
  //write function or logic  
}, []);
```

```
import { useEffect } from 'react';
```

```
const [old, setOld] = useState(false);
```

```
useEffect(() => {  
  //write function or logic  
}, [old]);
```


03.H

Form Handling

```
import { useState } from 'react';

function FormNama() {
  const [nama, setNama] = useState('');

  const handleChange = (e) => {
    setNama(e.target.value);
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    alert(`Halo, ${nama}!!`);
  };

  return (
    <form onSubmit={handleSubmit}>
      <input type="text" value={nama} onChange={handleChange} placeholder="Masukkan nama" />
      <button type="submit">Submit</button>
    </form>
  );
}
```

03.1

List Rendering

```
function App() {  
  const names = ["Deo", "Jason", "Budi"];  
  return (  
    <div>  
      {  
        names.map((name, index) => (  
          <Greeting name={name} />  
        ))  
      }  
    </div>  
  );  
}
```

04

Tailwind

Install



<https://tailwindcss.com/docs/installation/using-vite>

```
<div
  className="bg-white p-6 rounded-lg shadow-md w-full max-w-sm text-center cursor-pointer hover:shadow-lg
  transition"
  onClick={onClick}
>
  <h3 className="font-bold text-red-600 text-lg">{"MBG"}</h3>
  <p className="text-gray-600 text-sm mt-2">{"Biar anak sekolah makan enak"}</p>
  <p className="text-gray-500 text-xs mt-2">
    <a
      href={`https://sepolia-blockscout.lisk.com/address/${addressPIC}`}
      target="_blank"
      rel="noopener noreferrer"
    >
      {addressPIC}
    </a>
  </p>
</div>
```

MBG

Biar anak sekolah makan enak

0x1RGT7154748V1CE5176479CDGB574AE244B939B5

05

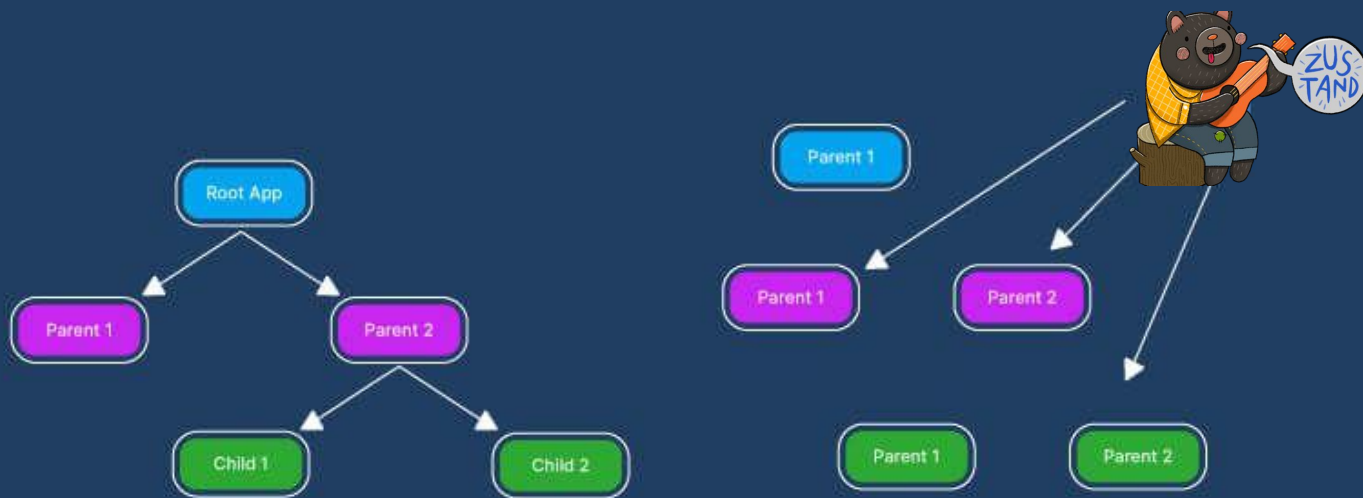
Zustand

Install



<https://zustand.docs.pmnd.rs/getting-started/introduction>

Problem



```
import { create } from 'zustand'

const useStore = create((set) => ({
  bears: 0,
  increasePopulation: () => set((state) => ({ bears: state.bears + 1 })),
  removeAllBears: () => set({ bears: 0 }),
  updateBears: (newBears) => set({ bears: newBears }),
}))

//-----

function BearCounter() {
  const bears = useStore((state) => state.bears)
  return <h1>{bears} bears around here...</h1>
}
```

```
function Controls() {  
  const increasePopulation = useStore((state) => state.increasePopulation)  
  const removeAllBears = useStore((state) => state.removeAllBears)  
  const updateBears = useStore((state) => state.updateBears)  
  
  return (  
    <div>  
      <button onClick={increasePopulation}>One up</button>  
      <button onClick={removeAllBears}>Remove all</button>  
      <button onClick={() => updateBears(10)}>Set to 10</button>  
    </div>  
  )  
}
```

Code Without Zustand

```
function App() {  
  const [bears, setBears] = useState(0);  
  
  return <Zoo bears={bears} setBears={setBears} />;  
}  
  
function Zoo({ bears, setBears }) {  
  return <BearCounter bears={bears} setBears={setBears} />;  
}  
  
function BearCounter({ bears, setBears }) {  
  return (  
    <div>  
      <h1>{bears} bears around here...</h1>  
      <button onClick={() => setBears(bears + 1)}>Tambah</button>  
    </div>  
  );  
}
```

Code With Zustand

```
import { create } from 'zustand';

const useBearStore = create((set) => ({
  bears: 0,
  increase: () => set((state) => ({ bears: state.bears + 1 })),
}));

export default useBearStore;
```

Code With Zustand

```
import Zoo from './Zoo';

function App() {
  return <Zoo />;
}

// Zoo.jsx

import BearCounter from './BearCounter';

function Zoo() {
  return (
    <div>
      <h2>Kebun Binatang</h2>
      <BearCounter />
    </div>
  );
}
```

Code With Zustand

```
// BearCounter.jsx

import useBearStore from './store/bearStore';

function BearCounter() {

  const bears = useBearStore((state) => state.bears);
  const increase = useBearStore((state) => state.increase);

  return (
    <div>
      <h1>{bears} bears around here...</h1>
      <button onClick={increase}>Tambah</button>
    </div>
  );
}
```

05

Navigasi dan Routing di React

Install



<https://reactrouter.com/>

```
import React from "react";
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";
import Home from "../views/Home";
import AdminPage from "../views/AdminPage";

const App: React.FC = () => {
  return (
    <Router>
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/admin" element={<AdminPage />} />
      </Routes>
    </Router>
  );
};

export default App;
```

Code

```
import { Link } from 'react-router-dom';

function Navbar() {
  return (
    <nav className="space-x-4">
      <Link to="/">Beranda</Link>
      <Link to="/admin">Admin</Link>
    </nav>
  );
}
```

```
import { useNavigate } from 'react-router-dom';

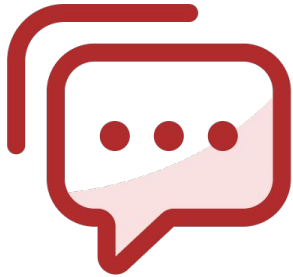
function LoginButton() {
  const navigate = useNavigate();

  const handleToAdmin = () => {
    // ...validasi login
    navigate('/admin');
  };

  return <button onClick={handleToAdmin}>Login</button>;
}
```

Hands On

Q & A



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Wrap Up

Share 3 things you learn today

On the Zoom Chat

Foto Bersama

Thank You



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